

Report
Of
AI RESUME ANALYZER AND JOB COMPATIBILITY SYSTEM

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By
Roll Number of Student: 2410030492
Name of Student: SAANVI SAXENA
Batch: 2024-2028

2026-27

CANDIDATE'S DECLARATION

I, SAANVI SAXENA, do hereby declare that the internship report titled “AI Resume Analyzer and Job Compatibility System” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: SAANVI SAXENA

Roll No.: 2410030492

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I extend my thanks to EduSkills and the AICTE-EduSkills Virtual Internship Program for providing the structured learning material and assessments in the domain of AI Deployment & Automation, and to the faculty of the School of Computer Science and Engineering, IILM University, Greater Noida, for their guidance during the preparation of this report.

I express my gratitude to my parents for their blessings.

Signature

Student Name: SAANVI SAXENA

Roll No.: 2410030492

INTERNSHIP COMPLETION CERTIFICATE

CERTIFICATE
OF INTERNSHIP







PRESENTED TO

Saanvi Saxena

has successfully achieved the Internship Credential by completing a 8-week program in the domain of

AI Deployment & Automation

During the internship, the learner demonstrated proficiency in:

- Foundations of AI Deployment & Production Readiness & Advanced API Deployment with FastAPI
- Containerization with Docker & Introduction to MLOps & CI/CD Pipelines
- Cloud Deployment Basics (AWS SageMaker, Google Vertex AI, Azure ML) & Deploying Generative AI Applications
- Building Advanced AI Agents with LangChain & LlamaIndex & AI-Powered Automation Workflows
- Integrating AI into Web/Mobile Apps & Security and Compliance in AI Deployment
- Container Orchestration with Kubernetes & Monitoring, Logging & Observability
- Performance Optimization & Cost Management & Model Drift Detection & Retraining & Advanced RAG & Agentic Workflows
- Case Studies & Best Practices in AI Deployment & Project: Deploy a Full-Stack RAG Application & Project: AI Agent for Business Automation & Capstone Project: End-to-End AI Solution for Business Problem

This certification recognizes the successful completion of the internship requirements.



Issued on: Aug 27, 2026
Certificate ID: 2026-595A888026



Mhu Swain
General Manager
(Learning & Development)
EduSkills

Internship Completion Certificate, EduSkills, Certificate ID 2026-595A888026, issued 27 August 2026

INTERNSHIP OFFER LETTER



Ref No: C17Y2026M081524624

Internship Offer Letter

Student Name : Saanvi Saxena
Stream : B.Tech
Branch : B.Tech(Computer Science and Engineering)
Institute Name : IILM University, Greater Noida
Internship Domain : AI Deployment & Automation
Internship Duration : June 2026 - August 2026
AICTE Student ID : STU699828db808d31771579611

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

Internship Offer Letter, EduSkills, Ref. No. C17Y2026M081524624

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CHAPTER 4

PROJECT DESCRIPTION

4.1 Introduction

This report describes the work carried out during the eight-week AICTE-EduSkills Virtual Internship in AI Deployment & Automation, conducted from July 2026 to August 2026. The internship dealt with what happens to an AI application after it has been written: how it is served through an API, packaged into a container, connected to other systems through automation workflows, and monitored once it is running.

The project built during this internship is the AI Resume Analyzer and Job Compatibility System. A user uploads a resume. The system reads it, collects real job listings from the web, and works out how closely the candidate matches each one, returning a compatibility score together with the skills the candidate already has, the skills the listings ask for that are missing, and what would be most useful to learn next. The aim is to give the candidate something they can act on rather than a number on its own.

4.2 Organization Profile

EduSkills is a non-profit organisation that works with colleges and industry partners to run virtual internship programmes for engineering students in India. This internship was conducted under the AICTE-EduSkills Virtual Internship Program in the domain of AI Deployment & Automation (Reference No. C17Y2026M081524624), and was completed on 27 August 2026 (Certificate ID 2026-595A888026).

The programme was delivered as a structured, week-wise learning plan with weekly assessments, project submissions and a final assessment test. It covered model serving with FastAPI, containerisation with Docker, MLOps and CI/CD, cloud deployment, generative AI serving, AI agents, retrieval-augmented generation, automation workflows, security, Kubernetes and monitoring. The knowledge gained from the programme was then applied to design and build the project described in this report.

4.3 Problem Statement

A candidate looking for a job runs into a simple practical limit: there are far more openings than any one person can go through. Every listing carries its own job

description, and checking each one against your own resume, skill by skill, is something you can do for five jobs but not for five hundred. Most candidates therefore apply on the strength of the job title alone, without really knowing whether they meet what the listing asks for.

The same problem appears in reverse once an application is rejected. A candidate is rarely told which requirement they failed to meet, so they do not know which skill is actually holding them back. Tools that claim to check a resume usually return a single percentage without saying which requirement produced it, which is not something anyone can act on.

This project addresses that gap. It reads the job descriptions on the candidate's behalf, compares them with the resume, and reports plainly which skills match, which are missing and what is worth learning next. It does not try to predict whether the candidate will be hired; that depends on interviews, competition and other factors the system cannot see.

4.4 Project Objectives

1. To accept a resume in PDF, DOC or DOCX format and check that it can be read.
2. To read the resume and build a structured profile of the candidate's skills, education and experience.
3. To collect real job listings from the web and identify what each listing requires.
4. To compare the candidate's profile with those requirements and produce a compatibility score.
5. To report matched skills, missing skills and related skills, and to suggest what the candidate should learn next.
6. To build and deploy the system using the deployment and automation methods taught during the internship.

4.5 Scope of the Project

The scope of this project is to analyse one resume against the job listings collected for it in a single run. The system is not limited to any single field: the candidate may be a software developer, a data analyst, a chartered accountant, a banker, a dentist or any other professional, and the same process is followed in every case, because the system works from whatever the resume states and whatever the job listing asks for. Resumes are accepted in PDF, DOC and DOCX form, and job listings are collected by scraping

job sites on the web. Within a run the system extracts the resume text, builds a candidate profile, reads the retrieved job descriptions, compares skills, experience and education, and returns a compatibility score with a skill breakdown and recommendations. The backend service is deployed as a container alongside the workflow.

Predicting hiring outcomes, rewriting resumes, applying to jobs on the candidate's behalf, screening candidates for employers, resumes written in languages other than English, and scanned resumes with no readable text are outside the scope of this project.

4.6 Technologies and Tools Used

Table 4.1 lists the tools used in the project and the reason each one is there. All of them are drawn from the topics covered during the internship.

Tool	Why it is used
n8n	Workflow orchestration tool. The AI workflow for this project is built in n8n: it takes the user's input, runs each stage of the system in order, passes data between the agents and the backend service, and returns the final result.
Large Language Model (LLM)	Reads and interprets text. Resumes and job descriptions are written in ordinary language, and a fixed keyword list cannot tell that “built a REST service in FastAPI” is evidence of a skill. The LLM is what makes the analysis agents possible.
FastAPI	Used for the small backend service that calculates the score. The score must come out the same every time for the same input, so it is worked out in code rather than by the model. FastAPI is chosen because it is a lightweight Python API framework that validates incoming data automatically, and because it is the framework taught during the internship.
Retrieval-Augmented Generation (RAG)	The extracted resume text is stored and indexed so that it can be searched during matching rather than passed around in full. The same store holds a small set of entries describing related technologies, which lets the system recognise that Flask is related to FastAPI, or that “sklearn” and “scikit-learn” are the same thing, so a candidate is given credit for skills that are close to what a listing asks for.
Docker	Packages the backend service together with everything it needs, so that it runs the same way on any machine and can sit alongside the workflow as a container.
Apify	A web scraping platform used to collect job listings from the internet. It runs a scraper against whichever job site is needed and returns the listings, with the full job description, for the roles being searched.

Table 4.1 Technologies and tools used in the project

4.7 System Architecture

The system is a single pipeline in which each stage does one job and passes its result to the next, as shown in Figure 4.1. The resume is uploaded, its text is extracted and stored, the stored text is analysed into a structured candidate profile, job listings are scraped and analysed for what they require, and the two are matched to produce a compatibility score. The candidate profile is what guides the search, so the listings collected are the

ones that suit the candidate's own background. A workflow layer holds the pipeline together, validating the uploaded file and running each stage in order.

Only the matching and scoring stage runs as a separate service. It is written with FastAPI and deployed in a Docker container, which keeps the calculation outside the language model so that the same resume and the same listing always produce the same score.

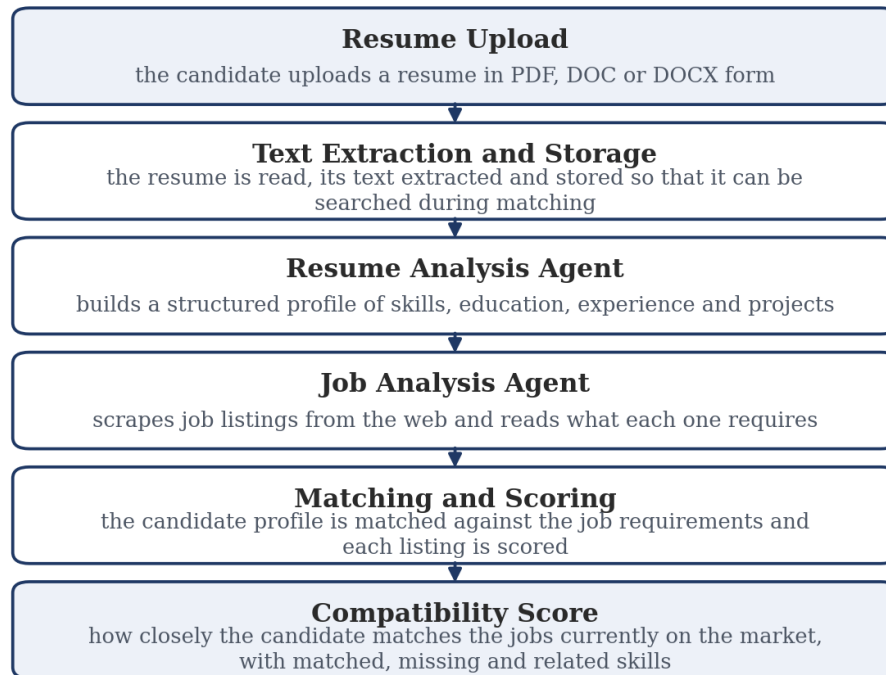


Figure 4.1 System architecture of the AI Resume Analyzer and Job Compatibility System

4.8 Methodology

The system works in five stages, summarised in Table 4.2.

Stage	What happens
1. Upload, extraction and storage	The user uploads a resume. The workflow checks that the file type is supported, and the resume is then converted into plain text and stored, so that it can be searched and referred back to during matching.
2. Resume analysis	The Resume Analysis Agent reads the stored text and produces a structured profile of the candidate: technical skills, tools, education, experience, projects and certifications. It records only what the resume actually states.
3. Job retrieval and analysis	Job listings are scraped from the web, guided by the roles the candidate profile points to. The Job Analysis Agent reads each description and lists what it asks for, separating the skills that are essential from those that are only preferred.
4. Matching and scoring	The candidate profile and the job requirements are compared, with the store used to recognise related technologies. A compatibility score is calculated, and matched, missing and related skills are listed.
5. Output	The result is returned to the user as a score, a match band, and a short set of recommendations on what to learn next.

Table 4.2 Stages of the system

How the score is worked out

A skill that the resume states and the listing requires counts in full. A skill the candidate does not have but for which they have a closely related one, such as Flask against a requirement for FastAPI, counts as half, and this is where the related-skill entries in the store are used. A requirement the candidate does not meet at all counts as nothing. Essential skills carry most of the weight, with smaller contributions from preferred skills, years of experience and education. The result is expressed as a percentage: 75 and above is reported as a strong match, 50 to 74 as a moderate match, and below 50 as a low match. The score describes how closely the resume lines up with what the listings ask for; it is not a prediction of whether the candidate will be hired.

Handling of failures and monitoring

Any stage can fail for reasons the system does not control: an unsupported file, a resume that is a scanned image with no readable text, a scraper that returns no listings, or a listing that states no requirements at all. In each of these cases the workflow stops at that point and reports what went wrong, instead of carrying on with incomplete information and producing a score that is not based on anything. The user therefore always receives a definite outcome: either a compatibility report, or a clear message explaining which stage failed and what to do about it, such as uploading the resume in a different format or running the analysis again once listings are available. Every run is recorded with its status and the stage it reached, and the backend service exposes a health check, so a failure can be traced without having to run the pipeline again.

Use of the internship learning

Table 4.3 shows where the topics taught during the internship are used in the project.

Internship topic	Where it is used in the project
API deployment with FastAPI	The scoring service
Containerisation with Docker	Packaging and running that service alongside the workflow
AI agents	The Resume Analysis Agent and the Job Analysis Agent
Retrieval-augmented generation	The store holding the extracted resume text and the related-skill entries
AI automation workflows	The workflow that runs the whole pipeline
Monitoring	Workflow execution records and a health check on the service

Table 4.3 Use of the internship learning in the project

Several topics formed part of the internship but are not used in this project, and are recorded here as learning rather than as project work: Kubernetes, cloud deployment

on AWS SageMaker, Google Vertex AI and Azure ML, CI/CD pipelines, and model drift detection and retraining. The project runs as a single service alongside the workflow and does not need them.

4.9 Expected Outcomes

For a given resume, the system returns a compatibility score for each listing analysed, the skills that matched, the skills that are missing, and a short list of what to learn next. A result might read, for example, as a moderate match of around seventy per cent, with Python, SQL and FastAPI matched, Docker and AWS missing, and containerisation identified as the most useful gap to close because it appears in most of the listings analysed. These figures illustrate the form of the output; they are not a measured result.

The value of the approach is that the number is explainable, because each part of it can be traced back to a specific requirement in a specific listing, and that it is consistent, because the comparison is calculated rather than generated. A candidate can therefore see not only how well they match, but why.

The system can only work from what the resume says, so a skill the candidate has but has not written down will be reported as missing. Job descriptions are often written loosely and mix genuine requirements with wishful ones, and the analysis inherits that imprecision. The range of fields the system can cover depends on the job sites the scraper is pointed at, so adding further sources widens the professions it can be used for. Future work would tailor the suggestions to individual listings, keep a history so that a candidate can see improvement over time, and move the service to a cloud deployment.

4.10 Certificates of Completion and Communication Proof

The internship completion certificate issued by EduSkills is reproduced below as proof of completion, followed by the communication received from EduSkills confirming completion of the programme. The internship offer letter is attached earlier in this report.



Internship Completion Certificate, EduSkills, Certificate ID 2026-595A888026, issued 27 August 2026

COMMUNICATION PROOF

From: EduSkills Foundation <noreply@eduskillsfoundation.org>
To: Saanvi Saxena <saanvi.saxena.cs28@iilm.edu>
Date: Thu, 27 Aug 2026 06:52:10 +0000
Subject: AICTE-EduSkills Internship Certificate download link

Dear Saanvi Saxena,

Congratulations!

You have successfully completed **10 weeks AI Deployment & Automation**. Your certificate is ready — please log in to your profile and download it.

Steps to download your certificate:

1. Login at: <https://eduskillsfoundation.org/login>
2. Go to the **Certificate Section** from your sidebar
3. Click **Download Certificate**

Please post your certificate on your LinkedIn profile to get better job opportunities and visibility from corporates.

Why You Should Share Your Achievements on LinkedIn

1. World's Largest Professional Platform

LinkedIn has 900+ million professionals and 60+ million companies worldwide.

2. Recruiters Actively Use LinkedIn

Around 8 out of 10 recruiters use LinkedIn to find and verify candidates.

3. Profiles Are Checked Before Hiring

More than 90% of hiring managers review LinkedIn profiles before shortlisting candidates.

4. Better Visibility for Freshers

Students who post certificates, internships, and projects appear in recruiter searches more often.

5. Skill-Based Hiring is Growing

Employers now focus more on skills, certifications, and practical experience.

6. Higher Chances of Job Opportunities

Active LinkedIn profiles with achievements get more internship and job opportunities.

Tip: Always share your certificates, tag EduSkills, mentors, and peers to increase visibility and professional networking.

Please Tag the following on LinkedIn:

@EduSkills Foundation | @AICTE | @AICTE NEAT | @Chandrasekhar Buddha | @Shyama Rath | @IILM University, Greater Noida

Hashtags:

#AICTEinternship #AICTE NEAT #AICTE #EduSkillsFoundation #AICTE_EduSkills_Internship #EduSkillsInternship #EduSkillsAcademy

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Best Regards,
EduSkills Team

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This is a system-generated email. Please do not reply to this message.

Completion communication received from EduSkills Foundation, 27 August 2026

CHAPTER 5 BIBLIOGRAPHY / REFERENCES

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